Lab 7 - Fuzz Testing Report

Train Fare | Hypothesis + Pytest | 12 November 2026

Objective: detect input crashes, improve validation, and verify fixes with fuzz tests.

1. Original application run

Baseline run of parse_pnr, journey_distance, and berth_type.

```
(.venv) ankithal@Ankithas-MacBook-Pro Lab 7 % python train_fare.py
==== Running parse_pnr ====
Enter a 10 digit PNR (e.g. 4521367890): 4521367890
Zone and Number: (4, 521367890)

==== Running journey_distance ====
Enter station distances as CSV (e.g. 0,120,340): 0,120,340
Total Distance: 340

==== Running berth_type ====
Enter coach and seat (e.g. S3-45): S3-45
Coach and Berth: ('S3', 'MB')
```

2. Fuzz tests: failures before fixes

Hypothesis identifies three failing test functions.

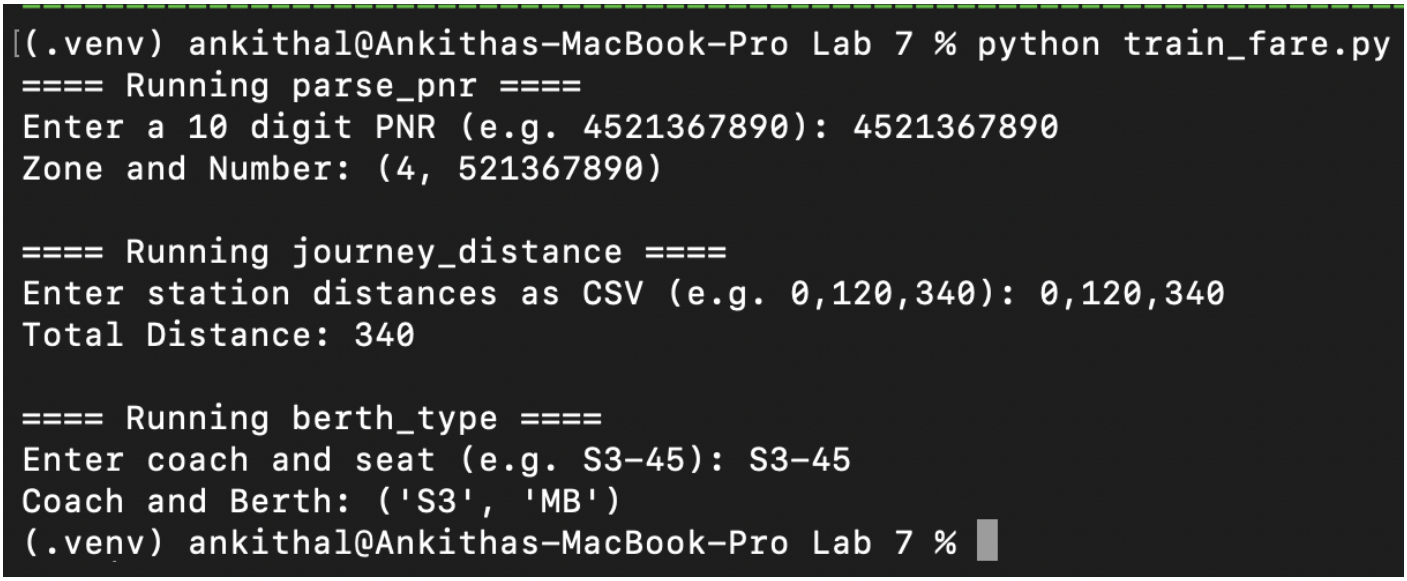
```
=========================================================================== 3 failed in 2.54s ============================================================================
(.venv) ankithal@Ankithas-MacBook-Pro Lab 7 % python -m pytest test_train_fare.py -v --tb=long
============================================================================= test session starts ==============================================================================
platform darwin -- Python 3.9.6, pytest-8.4.2, pluggy-1.6.0 -- /Users/ankithal/Desktop/L6_L7_25_train_fare/Lab 7/.venv/bin/python
cachedir: .pytest_cache
hypothesis profile 'default'
rootdir: /Users/ankithal/Desktop/L6_L7_25_train_fare/Lab 7
plugins: hypothesis-6.141.1
collected 3 items

test_train_fare.py::test_parse_pnr_no_crash FAILED                                                                                                                   [ 33%]
test_train_fare.py::test_journey_distance_no_crash FAILED                                                                                                            [ 66%]
test_train_fare.py::test_berth_type_no_crash FAILED                                                                                                                  [100%]

================================================================================== FAILURES ==================================================================================
_____________________________________________________________________________ test_parse_pnr_no_crash _____________________________________________________________________________
  + Exception Group Traceback (most recent call last):
  |   File "/Users/ankithal/Desktop/L6_L7_25_train_fare/Lab 7/test_train_fare.py", line 6, in test_parse_pnr_no_crash
  |     def test_parse_pnr_no_crash(s):
  |   File "/Users/ankithal/Desktop/L6_L7_25_train_fare/Lab 7/.venv/lib/python3.9/site-packages/hypothesis/core.py", line 2124, in wrapped_test
  |     raise the_error_hypothesis_found
  | exceptiongroup.ExceptionGroup: Hypothesis found 3 distinct failures. (3 sub-exceptions)
  +-+---------------- 1 ----------------
    | Traceback (most recent call last):
    |   File "/Users/ankithal/Desktop/L6_L7_25_train_fare/Lab 7/test_train_fare.py", line 8, in test_parse_pnr_no_crash
    |     parse_pnr(s)
    |   File "/Users/ankithal/Desktop/L6_L7_25_train_fare/Lab 7/train_fare.py", line 4, in parse_pnr
    |     return int(pnr[0]), int(pnr[1:])
    | ValueError: invalid literal for int() with base 10: ''
    |
    | During handling of the above exception, another exception occurred:
    |
    | Traceback (most recent call last):
    |   File "/Users/ankithal/Desktop/L6_L7_25_train_fare/Lab 7/test_train_fare.py", line 10, in test_parse_pnr_no_crash
    |     assert False, f"parse_pnr crashed with: {e}"
    | AssertionError: parse_pnr crashed with: invalid literal for int() with base 10: ''
    | assert False
    | Falsifying example: test_parse_pnr_no_crash(
    |     s='0',
    | )
    +---------------- 2 ----------------
    | Traceback (most recent call last):
    |   File "/Users/ankithal/Desktop/L6_L7_25_train_fare/Lab 7/test_train_fare.py", line 8, in test_parse_pnr_no_crash
    |     parse_pnr(s)
    |   File "/Users/ankithal/Desktop/L6_L7_25_train_fare/Lab 7/train_fare.py", line 4, in parse_pnr
    |     return int(pnr[0]), int(pnr[1:])
    | IndexError: string index out of range
    |
    | During handling of the above exception, another exception occurred:
    |
    | Traceback (most recent call last):
```

3. Fixed application run

Valid sample inputs still produce the expected outputs.



4. Fuzz tests: passing after fixes

All three fuzz tests pass.

